Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of Structural Design and Fabrication Process for ScAlN-Based mmWave BAW Devices			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (x), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		**Electrical · Electronics**	
	Medium Category	Small Category	Medium Category	Small Category
	ICT Modules and Components	Mobile Communication Modules and Components	Semiconductor Devices and Systems	Compound Semiconductor Devices
Concept of Proposed Technology	**"Acoustic Barrier" in Semiconductor Devices and Next-Gen Communications:** As wireless communication evolves from 5G to 6G and LEO (Low Earth Orbit) satellite networks, the operational frequency is shifting rapidly from Sub-6 GHz to the millimeter-wave (mmWave) spectrum, specifically targeting above the X-band(8-12GHz). However, conventional FBAR technology based on Aluminum Nitride (AlN) faces a physical limit known as the "Acoustic Barrier" at frequencies above 3GHz. **Challenge:** To resonate above X-band, piezoelectric films must become thin (<400nm), leading to severe degradation in **Q-factor** and **Power Handling** capabilities. Furthermore, traditional AlN lacks the electromechanical coupling coefficient (k_{eff}^2) required for the ultra-wide bandwidths of next-gen networks. **Solution:** The integration of **Scandium Aluminum Nitride (ScAlN)** is essential. ScAlN increases the piezoelectric response significantly, enabling wider bandwidths and higher efficiency without dangerously thinning the membrane.			

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Rationale for Joint R&D Project	By combining German material science with Korean manufacturing integration, this joint R&D aims to bypass the limitations of current market solutions. **Structural Innovation:** The collaboration will move beyond simple FBAR scaling. By utilizing the high acoustic velocity and coupling of the German ScAlN film, the Korean partner can implement advanced structures such as more robust membranes even at X-band. **Supply Chain Security:** The Korean partner's established relationship with demand-side clients ensures a direct feedback loop. This allows the consortium to rapidly iterate on the ScAlN film parameters based on actual device performance data (S-parameters, Power Durability) measured in the Korean facility.
Key Words	1. Piezoelectric thin film 2. Aluminum Scandium Nitride 3. Magnetron sputter deposition 4. Bulk Acoustic Wave (BAW) filter 5. Wireless communication

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	The **German partner** overcomes the material limits by providing high-k_{eff}^2 ScAlN thin film magnetron sputter technology, while the **Korean partner** overcomes the manufacturing limits by applying precision MEMS fabrication, frequency trimming, and packaging technologies to transform the material into a viable above 6GHz filter product for their existing customer base.			
R&D Target Specifications	ScAlN film with Sc concentration of 20–30%, while maintaining a Rocking Curve FWHM of <1.5°. BAW Filter with frequency ≥6GHz, coupling k_{eff}^2 ≥10% and Quality factor ≥ 1,000			
R&D Plan	WP1 Thin film process development (Fraunhofer FEP): Development of magnetron sputter technology for deposition of ScAlN films will be based on FEP's proprietary magnetron sputter sources for reactive sputtering at very competitive deposition rates of up to 3 nm/sec onto substrate size 8". Hardware and technology shall provide superior process control and adjustment of critical parameters such as morphology, crystallinity and mechanical stress. This technology will enable ScAlN films with pure c-axis orientation even at low film thickness (<400nm), low FWHM of rocking curve 1.5° and strong electromechanical coupling. Reaching this target is supported by the parallel development of magnetron sputter technology for deposition of suitable metallic electrode and seed layer. Characterization of films will be performed using XRD, AFM, PFM, Ellipsometry and ferroelectric testing. WP2 Development of structural design and process technology for BAW filter device fabrication (Korean partner): Enhancement of Q-factor performance through the development of etching processes and sacrificial layer processes for resonator formation. Evaluation of characteristics such as Q-factor and coupling coefficient at the BAW resonator wafer die level. WP3 Device integration (Korean partner):			

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BAW device packaging with small form factors, Research on improving electrical characteristics of filter devices, including S-parameters. Durability verification of the device and reliability enhancement testing. Field tests to see if technology is suitable for modularization and system integration by client companies applying the developed devices.

Work package	content	Year 1				Year 2				Year 3			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
WP 1	**Thin film proces development**												
WP 1.1	Pre-study												
WP 1.2	ScAlN technology development												
WP 1.3	Seed and electrode film development												
WP 1.4	Film deposition for devices												
WP 1.5	Film characterization												
WP 2	**BAW filter device fabrication**												
WP 3	**Device integration**												
Milestones			M1: Pre-study finished					M2: ScAlN properties					

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**German Partner FEP: Upstream Excellence in ScAlN Material Process** **Role:** The German partner FEP with its substantial experience in development of magnetron sputter sources and processes will provide the core technology for depositing high-quality **ScAlN thin films.** **Technical Necessity:** The performance of an above 6GHz BAW filter is entirely dependent on the crystallinity and doping uniformity of the piezoelectric layer. Achieving a **Sc concentration of 20–30%** while maintaining a Rocking Curve FWHM of <1.5° is a difficult material science challenge. The German partner's expertise in controlling stress and stoichiometry in ScAlN deposition solves the "input quality" problem that limits standard MEMS foundries.
Expected R&R of Korean SME Partner	**Korean Partner: Downstream Excellence in MEMS Fabrication & Commercialization** ◦ **Role:** The Korean partner possesses established **MEMS device fabrication** capabilities (Air Cavity formation, Electrode patterning) and mass-production infrastructure for **filters** currently supplied to client companies. ◦ **Technical Necessity:** A high-quality film alone does not make a filter. The Korean partner will execute the complex **Front-End WBS** (cavity release, lithography) and critical **Wafer-Level Post-Processing** tasks such as **Frequency Trimming** (Ion Beam Milling) to ensure precise frequency targeting at above 6GHz. Furthermore, their existing qualification framework (AEC-Q200/JEDEC reliability testing) and **Wafer Level Packaging (Micro-capping)** technology ensure the device can be integrated into client modules immediately.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Market Penetration:** This project will result in a commercial-grade filter device, directly addressing the supply shortage for **LEO satellite user terminals** and **6G infrastructure**, a market currently dominated by bulky ceramic or waveguide filters. **Technological Independence:** Establishing a proprietary supply chain for ScAlN-based mmWave devices reduces reliance on dominant IDMs (e.g., Broadcom, Qorvo) and positions both nations as leaders in the high-frequency RF component ecosystem.

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6. References

(1) Juhun Baek, Stephan Barth, Tom Schreiber, Hagen Bartzsch, John Duncan, Gianluca Piazza: 51.3 GHz Overmoded Bulk Acoustic Resonator Using 35% Scandium Doped Aluminum Nitride, IEEE, 2025, JOURNAL OF MICROELECTROMECHANICAL SYSTEMS